Zomato Bangalore Restaurant Analysis - Complete EDA Project Guide

1. Import Libraries

Purpose: Import required libraries for data manipulation and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

2. Upload Dataset

Purpose: Upload the Zomato dataset into Google Colab.

```
from google.colab import files
uploaded = files.upload()
```

3. Load Dataset

Purpose: Read the CSV file into a Pandas DataFrame.

```
df = pd.read_csv("zomato.csv")
```

4. Explore Data

Purpose: Understand dataset structure, rows, columns, and data types.

```
df.head()
df.shape
df.columns
df.info()
```

5. Missing Values Analysis

Purpose: Identify missing values present in the dataset.

```
df.isnull().sum()
```

6. Remove Unnecessary Columns

Purpose: Remove URL and phone columns that are not useful for analysis.

```
df.drop(['url','phone'], axis=1, inplace=True)
```

7. Remove Duplicate Records

Purpose: Remove repeated restaurant entries.

```
df.duplicated().sum()
df.drop_duplicates(inplace=True)
```

8. Clean Ratings Column

Purpose: Convert ratings into numeric format.

```
df['rate'] = df['rate'].replace(['NEW','-'], np.nan)
df['rate'] = df['rate'].str.split('/').str[0]
df['rate'] = df['rate'].astype(float)
```

9. Clean Cost Column

Purpose: Remove commas and convert cost values into numeric format.

```
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].astype(str)
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].str.replace(',', '')
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].astype(float)
```

10. Rating Distribution Analysis

Purpose: Understand how restaurant ratings are distributed.

```
plt.figure(figsize=(10,5))
sns.histplot(df['rate'], bins=20, kde=True)
plt.title("Restaurant Ratings Distribution")
plt.show()
```

11. Online Ordering Analysis

Purpose: Analyze availability of online ordering.

```
sns.countplot(x='online_order', data=df)
plt.title("Online Ordering Availability")
plt.show()
```

12. Table Booking Analysis

Purpose: Analyze table booking services offered by restaurants.

```
sns.countplot(x='book_table', data=df)
plt.title("Table Booking Availability")
plt.show()
```

13. Top Restaurant Locations

Purpose: Find areas with the highest restaurant density.

```
top_locations = df['location'].value_counts().head(10)
top_locations.plot(kind='bar')
plt.title("Top 10 Restaurant Locations")
plt.show()
```

14. Top Cuisines Analysis

Purpose: Discover the most popular cuisines.

```
cuisine = df['cuisines'].dropna().str.split(',').explode()
cuisine.value_counts().head(10).plot(kind='bar')
plt.title("Top Cuisines")
plt.show()
```

15. Cost Distribution Analysis

Purpose: Understand pricing patterns across restaurants.

```
sns.histplot(df['approx_cost(for two people)'], bins=40)
plt.title("Cost Distribution")
plt.show()
```

16. Rating vs Cost

Purpose: Check whether expensive restaurants receive better ratings.

```
sns.scatterplot(
    x='approx_cost(for two people)',
    y='rate',
    data=df
)
```

```
plt.show()
```

17. Correlation Heatmap

Purpose: Analyze relationships between numerical variables.

```
numeric_df = df.select_dtypes(include=np.number)
sns.heatmap(numeric_df.corr(), annot=True)
plt.show()
```

18. Final Business Insights

Purpose: Summarize findings.

```
1. Most ratings lie between 3.5 and 4.2
2. Online ordering is highly popular.
3. Quick Bites restaurants dominate the market.
4. North Indian cuisine is widely preferred.
5. Cost does not strongly influence ratings.
6. Koramangala, BTM, Whitefield and Indiranagar have many restaurants.
7. Table booking restaurants generally receive higher ratings.
```